Problem Statement – Practical 1

Student Profile Viewer App – A Basic Android Application Using Activities, Services, and User Interface Components

You are required to design and develop an Android application named **Student Profile Viewer**. The objective of this practical is to help students understand the basic components of Android such as **Activities**, **Services**, **Content Providers**, basic **User Interface elements**, and working with an **Android Virtual Device (AVD)** and **USB Debugging**.

Procedure:

Open Android Studio.

Click File → New → New Project.

Select Empty Views Activity.

Name: StudentProfileViewer

Package Name: package com.example.studentprofileviewer

Language: Java

Minimum SDK: API 21 (Android 5.0)

Click Finish.

Source Code

Design the Main Layout (activity_main.xml)

Procedure

1. In **Project Explorer**, go to:
   app → res → layout → activity_main.xml.
2. Click **Code tab** (bottom of editor).
3. Replace content with:

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:fillViewport="true">
    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="vertical"
        android:padding="16dp">
        <!-- Student Name -->
        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
```

```
    android:text="Student Name"
    android:textStyle="bold"
    android:paddingTop="8dp"/>

<EditText
    android:id="@+id/etName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Name"
    android:inputType="textPersonName"/>
<!-- Roll Number -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Roll Number"
    android:textStyle="bold"
    android:paddingTop="8dp"/>
<EditText
    android:id="@+id/etRoll"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Roll Number"
    android:inputType="number"/>
<!-- Course -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Course"
    android:textStyle="bold"
    android:paddingTop="8dp"/>
<EditText
    android:id="@+id/etCourse"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter Course"
    android:inputType="text"/>
<!-- Email -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Email"
    android:textStyle="bold"
    android:paddingTop="8dp"/>
<EditText
    android:id="@+id/etEmail"
    android:layout_width="match_parent"
```

```
        android:layout_height="wrap_content"
        android:hint="Enter Email"
        android:inputType="textEmailAddress"/>
    <!-- Button -->
    <Button
        android:id="@+id/btnViewProfile"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="View Profile"
        android:paddingTop="12dp"
        android:layout_marginTop="16dp"/>
  </LinearLayout>
</ScrollView>
```

Create Profile Activity

Procedure

1. Right-click app → java → com.example.studentprofileviewer.
2. Select **New → Activity → Empty Activity**.
3. Name: ProfileActivity.
4. Click **Finish**.

Android Studio generates ProfileActivity.java and activity_profile.xml

Layout (activity_profile.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="16dp">
  <TextView
    android:id="@+id/tvName"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="18sp"
    android:textStyle="bold"
    android:padding="8dp"/>
  <TextView
    android:id="@+id/tvRoll"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:textSize="18sp"
    android:padding="8dp"/>
```

```
    <TextView
        android:id="@+id/tvCourse"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:padding="8dp"/>
    <TextView
        android:id="@+id/tvEmail"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:textSize="18sp"
        android:padding="8dp"/>
</LinearLayout>
```

Code (<u>ProfileActivity.java</u>)

```
package com.example.studentprofileviewer;
import android.content.Intent;
import android.os.Bundle;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class ProfileActivity extends AppCompatActivity {
    TextView tvName, tvRoll, tvCourse, tvEmail;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_profile);
        // Initialize TextViews
        tvName = findViewById(R.id.tvName);
        tvRoll = findViewById(R.id.tvRoll);
        tvCourse = findViewById(R.id.tvCourse);
        tvEmail = findViewById(R.id.tvEmail);
        // Get data from Intent (sent from MainActivity)
        String name = getIntent().getStringExtra("NAME");
        String roll = getIntent().getStringExtra("ROLL");
        String course = getIntent().getStringExtra("COURSE");
        String email = getIntent().getStringExtra("EMAIL");
        // Set data to TextViews
        tvName.setText("Name: " + name);
        tvRoll.setText("Roll Number: " + roll);
        tvCourse.setText("Course: " + course);
        tvEmail.setText("Email: " + email);
        // Start LoggingService in background
        Intent serviceIntent = new Intent(this, LoggingService.class);
        serviceIntent.putExtra("NAME", name);
        serviceIntent.putExtra("ROLL", roll);
```

```
        startService(serviceIntent);
    }
}
```

Add Logging Service

Procedure

1. Right-click app → java → com.example.studentprofileviewer.
2. Select **New → Java Class**
3. Name: LoggingService.
4. Click **Finish**.

```
package com.example.studentprofileviewer;

import android.app.Service;
import android.content.Intent;
import android.os.IBinder;
import android.util.Log;

public class LoggingService extends Service {

    private static final String TAG = "LoggingService";

    @Override
    public void onCreate() {
        super.onCreate();
        Log.d(TAG, "Service Created");
    }

    @Override
    public int onStartCommand(Intent intent, int flags, int startId) {
        // Get data passed from ProfileActivity
        String name = intent.getStringExtra("NAME");
        String roll = intent.getStringExtra("ROLL");

        Log.d(TAG, "Profile Viewed: " + name + " (Roll: " + roll + ")");

        // Service runs once and stops (not sticky)
        return START_NOT_STICKY;
    }

    @Override
    public void onDestroy() {
        super.onDestroy();
        Log.d(TAG, "Service Destroyed");
    }
```

```
    @Override
    public IBinder onBind(Intent intent) {
        // Not using bound service here
        return null;
    }
}
```

MainActivity.java

```
package com.example.studentprofileviewer;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;
import android.widget.EditText;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    EditText etName, etRoll, etCourse, etEmail;
    Button btnViewProfile;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        etName = findViewById(R.id.etName);
        etRoll = findViewById(R.id.etRoll);
        etCourse = findViewById(R.id.etCourse);
        etEmail = findViewById(R.id.etEmail);
        btnViewProfile = findViewById(R.id.btnViewProfile);

        btnViewProfile.setOnClickListener(v -> {
            Intent intent = new Intent(MainActivity.this, ProfileActivity.class);
            intent.putExtra("NAME", etName.getText().toString());
            intent.putExtra("ROLL", etRoll.getText().toString());
            intent.putExtra("COURSE", etCourse.getText().toString());
            intent.putExtra("EMAIL", etEmail.getText().toString());
            startActivity(intent);
        });
    }
}
```

Register Service in Manifest

Open AndroidManifest.xml → inside <application> add:

<service android:name=".LoggingService" />

Run and Test

Procedure

1. Connect your phone via USB (enable **Developer Options → USB Debugging**).
2. In Android Studio, click **Run** .
3. Select your device.
4. Enter student details → click **View Profile**.
5. ProfileActivity opens and shows details.
6. Open **Logcat** (bottom panel in Android Studio).

   Filter by LoggingService.

   You'll see:

   D/LoggingService: Service Created

   D/LoggingService: Profile Viewed: Ramesh (Roll: 101)

D/LoggingService: Service Destroyed